

Malhar Tushar Patwardhan

+91-9405747208 | f20240502@pilani.bits-pilani.ac.in | [LinkedIn](#) | [GitHub](#)



EDUCATION

Birla Institute of Technology and Science, Pilani

B.E. Mechanical Engineering - CGPA: 5.78/10

Pilani, Rajasthan

Aug 2024 - Present

Global Public School and Jr. College / Wisdom High International School

MSBSHSE - 78.83% (XIIth) | ICSE - 94.33% (Xth)

Maharashtra

2021 - 2023

RELEVANT TECHNICAL SKILLS

Technical Proficiency - C/C++, Python, ANSYS Fluent, MATLAB, XFLR5, OptimumLap, SolidWorks, AutoCAD, Fusion 360, Autodesk CFD Ultimate, COMSOL, SQL, LaTeX

Subjects/Electives - Fluid Dynamics, Data Mining in Mechanical Sciences, Numerical Fluid Flow and Heat Transfer, Manufacturing Processes, Heat Transfer, Material Sciences, Mechanics of Solids

WORK EXPERIENCE/INTERNSHIPS

Structural Steel & Construction Management Intern

May 2026 - Jul 2026

Kalpataru Projects International Ltd

Bagdogra Airport Civil Enclave, West Bengal, India

- Independently designed and deployed a **Python/VLM (Qwen2.5-VL via Ollama)** automation script to extract structured tabular data from **1500** fabrication drawing screenshots into Excel; a self-initiated solution to a manual transcription bottleneck, not a formally assigned task.
- Performed **Dye Penetrant Testing** and paint thickness (DFT) testing on fabricated structural steel; observed a formal welder qualification test with external laboratory bend/tensile verification.
- Studied **Autodesk Revit** BIM models for HVAC and firefighting piping coordination and clash detection; reviewed heat load calculations and chiller/AHU sizing methodology.
- Observed Air Handling Unit (AHU) assembly and ventilation ducting installation; studied grooved pipe coupling systems used across HVAC and firefighting piping networks.

PROJECTS

Automated Engineering Drawing Digitisation Pipeline

Self-Initiated, Kalpataru Projects International Ltd

2026

- Built a Python automation pipeline using a vision-language model (**Qwen2.5-VL** via Ollama) to parse tabular data from AutoCAD-style fabrication drawing screenshots and write structured output to Excel using **openpyxl**.
- Processed **1500** fabrication drawing screenshots, extracting all **7** component fields per drawing.
- Eliminated manual transcription and its associated errors, feeding directly into **4** master AutoCAD weight annotations used for tower crane positioning and lift planning.

Formula Bharat Rear Wing Aerodynamic Design

Team Inspired Karters Electric, BITS Pilani

2026 – Present

- Selected airfoils and designed a multi-element rear wing, running ANSYS Fluent CFD and XFLR5 simulations across geometry variables (slot gap, overlap, flap deflection, angle of attack).
- Evaluated candidates from a **113**-airfoil comparison spreadsheet and ran **1000+** CFD iterations to explore the design space.
- Optimised the downforce-to-drag trade-off for the final wing configuration used on the team's EV race car.

RESEARCH, PUBLICATIONS & CERTIFICATIONS

Co-presented Paper - "The Impact of Bleed and Bleedless Air Systems on Small Turbojet Engines Used in UAVs," Product Lifecycle Modelling, Simulation and Synthesis workshop, Dept. of Mechanical Engineering, BITS Pilani (Nov 2025).

Certifications - MATLAB Onramp; Machine Learning Onramp (MathWorks Training Services); Certified Solidworks Associate.

POSITIONS OF RESPONSIBILITY

Aerodynamics Lead

Team Inspired Karters Electric (Formula Bharat), BITS Pilani

Jun 2026 - Present

- Lead engineer for design, simulation, and analysis of all aerodynamic components on the team's EV race car
- Designed a multi-element rear wing from scratch, running **1000+** ANSYS Fluent/XFLR5 iterations varying airfoil, angle of attack, and element spacing

Secretary

Mechanical Engineering Association, BITS Pilani

Jun 2026 - Present

- Coordinate departmental communication and event operations for the Mechanical Engineering Association.

OASIS Coordinator and APOGEE Joint Coordinator

Mechanical Engineering Association, BITS Pilani

Aug 2025 - Mar 2026

- Conceptualized, pitched, and executed **5** major technical events across BITS Pilani's flagship fests
- Managed operational logistics for a premier sponsored technical event carrying an **INR 50,000** prize pool.

ACHIEVEMENTS & ACTIVITIES

Skating & Athletics - Member, BITS Pilani skating team; competed in BOSM '24, '25 and InterBITS 2025, winning **3 gold, 3 silver, and 1 bronze** medal; also a member of the long-distance running team.

Peer Mentorship Program - Peer Mentor, guiding **6** juniors through academics, sports, and campus life (Jul 2025 - Apr 2026).